

Quantitative Trading and Price Impact

End-to-end pipeline with non-parametric impact
and multi-day carry backtest

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*Modular src/price_impact/ package | thin orchestration notebook | every backtest run writes to
saved/<name>/*

Outline

What we will cover

1. Data preparation §2.1
2. Impact model fitting: OW, AFS, non-parametric extension §2.2
3. Backtest engine (Waelbroeck simulator) §2.3
4. Synthetic alphas §2.4
5. Optimal trading strategy §2.5
6. Performance metrics and TCA decomposition §2.6
7. Sensitivity analysis and stress testing §2.7

Panel

- 2019 US equities, 10-s bins (monthly CSVs ~200 MB each)
- **Top-20 stocks** by full-year total $|q_t|$
- Final panel: ~11.0M bin rows, 250 trading days

Daily statistics (trailing 20-day rolling)

$$\sigma_{i,d}^2 = \overline{\text{Var}_t r_{i,d,t}}, \quad \text{ADV}_{i,d} = \overline{\sum_t |q_{i,d,t}|}$$

Used to normalise flow so impact coefficients are comparable across stocks and days.

In-sample / out-of-sample protocol

10 rolling windows:

- train on month m
- tune γ on $m+1$ (NP only)
- validate on $m+2$

for $m = 1, \dots, 10$.

```
build_panel(data_dir, year,
            top_n=20,
            lookback_days=20)
→ PanelData(bins, daily_stats,
            stocks)
```

Module data = 165 lines, deterministic, no I/O outside the bin CSVs.

Both models build the OU recursion

$$\bar{l}_{t+1} = (1 - \beta)\bar{l}_t + \tilde{q}_t, \quad \beta = \frac{\ln 2}{H_{\text{bins}}}.$$

They differ only in the **flow kernel** – how a child trade of size q_t pushes the state $\bar{l}$:

OW: $\tilde{q}_t = \sigma q_t / \text{ADV}$ (linear)

AFS: $\tilde{q}_t = \sigma \text{sgn}(q_t) \sqrt{|q_t| / \text{ADV}}$ (concave)

Why AFS is interesting. The model lets us *specify* the flow kernel $f(q)$ rather than inheriting the linear form by default. We pick the $\sqrt{q/V}$ because it lines up with the empirical $\sqrt{Q/V}$ meta-order law – mechanism explored on *slide 9* (Backtest engine), where the kernel choice becomes the operative modelling decision.

Optimisation problem. Joint

$$\max_{H, \alpha, \lambda} R_{\text{OOS}}^2(H, \alpha, \lambda).$$

For fixed H , OLS pins $(\hat{\alpha}, \hat{\lambda})$ in closed form, so the problem reduces to a **1-D scan over H** – the half-life that yields the best-fitting impact regression:

$$\max_H R_{\text{OOS}}^2(H, \hat{\alpha}(H), \hat{\lambda}(H)).$$

Regression target. At horizon $\tau = 6$ bins (60 s),

$$y_t = \alpha + \lambda x_t + \varepsilon_t,$$

with $x_t = \bar{l}_t - \bar{l}_{t-\tau}$, $y_t = (P_t - P_{t-\tau})/P_{t-\tau}$. Fit via additive sufficient statistics $(S_x, S_y, S_{xx}, S_{yy}, S_{xy}, n)$ – enables fast rolling re-fits and the H -grid sweep on the next slide.

Half-life grid search

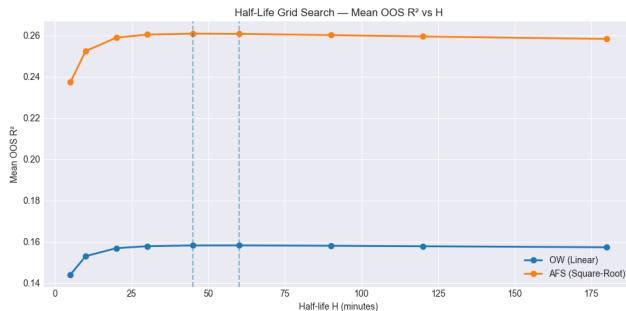
§2.2

H is a market property, not a model parameter

Mean OOS R^2 over
 $H \in \{5, 10, 20, 30, 45, 60, 90, 120, 180\}$
min:

H (min)	OW	AFS
10	0.1530	0.2525
30	0.1579	0.2604
45	0.1582	0.2609
60	0.1583	0.2608
90	0.1581	0.2602
120	0.1578	0.2595

OW peaks at $H=60$, AFS at $H=45$ – but not separate truths.



Why the midpoint? $H^* = 52.5$ min

- H is a **market property** – liquidity recovery rate. It should not depend on whether we model with OW or AFS flow
- Cross-model agreement is the test: both prefer 45–60 min within $< 10^{-4} R^2$. Midpoint is the natural single estimate of one universal market parameter

Non-parametric extension

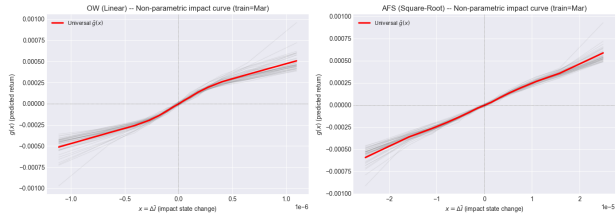
Method + estimated functional form

ENHANCEMENT #1

Replace the parametric $y = \alpha + \lambda x$ with a binned $g(x)$, applied to *both* OW and AFS (same procedure, different $x_t = \Delta \bar{I}$):

$$g_{i,b}^{\text{reg}}(\gamma) = \frac{S_{y,i,b} + \gamma \bar{g}_b}{n_{i,b} + \gamma}.$$

- $B = 15$ quantile bins on train month m (pooled across stocks)
- $\bar{g}_b$ = pooled universal curve across all stocks
- Per-stock $\hat{g}_{i,b}$ shrunk toward $\bar{g}_b$ with strength γ
- γ tuned on test month $m+1$, validated on $m+2$



Estimated $g(x)$ at train month $m=3$. Grey: per-stock $\hat{g}_i(\cdot)$; dashed: universal $\bar{g}(\cdot)$.

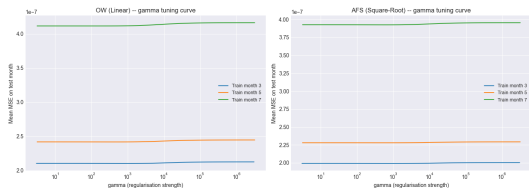
Left (**OW**, $x = \sigma \Delta \bar{I}_{lin}$): curves cluster around an approximately linear shape – but the model forces linearity, so the question is how much that constraint costs.

Right (**AFS**, $x = \sigma \Delta \bar{I}_{sqrt}$): curves are visibly **concave**, consistent with the empirical square-root impact regularity.

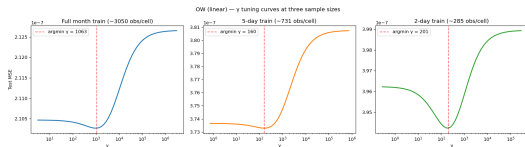
γ regularisation – regime-correctness

Flat in our data; kicks in for shorter training windows

Headline (1 month, ~ 3000 obs/cell)



Stress (train window 1 mo $\rightarrow$ 5 d $\rightarrow$ 2 d)



Test-month MSE vs γ , across multiple windows.
Curves **flat across 6 orders of magnitude** –
per-stock estimators are already precise; shrinkage
has nothing to fix.

Train window	Med. n	OW gain	γ^*/n
1 month	~ 3000	+0.0005	0.16
5 days	~ 720	+0.0025	0.56
2 days	~ 290	+0.0070	0.93

Estimator is **regime-correct**: trusts per-stock data when there's plenty, pools toward universal when starved. OOS gain grows $\sim 14\times$ as data shrinks $\sim 10\times$; shrinkage strength rises $6\times$. Headline flatness is a property of the data-rich regime, not a defect.

Headline OOS R^2 – results

§2.2

Apples-to-apples comparison of parametric vs non-parametric

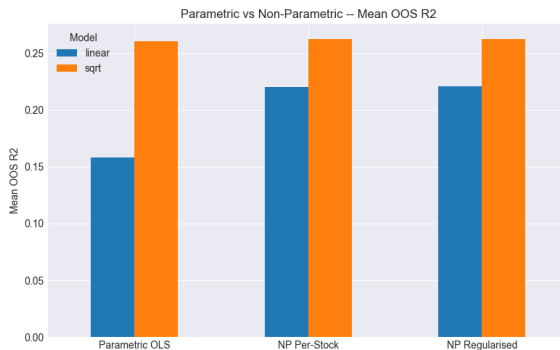
Pooled OOS R^2 (panel-wide null, same merged subset)

Model	Param.	NP Reg.	Δ
OW	0.1612	0.2206	+0.059
AFS	0.2689	0.2629	−0.006

Two findings

- **NP lifts OW by +5.9 pp** – linearity is a binding constraint
- **NP ties AFS** (−0.6 pp): the 1-parameter sqrt form sits on the efficient frontier; 15 parameters buy nothing

Aggregation matters: under per-stock R^2 averaging, NP appears to beat AFS by +0.2 pp – an artefact that disappears under pooled R^2



Mean OOS R^2 across rolling windows, by model and estimator. AFS dominates OW in every estimator family.

$$P_{\text{sim}}(t) = P_0(1 + \text{cumret}(t) + g_{\text{sim}}(t) - g_{\text{ref}}(t)), \quad g(t) = \lambda \bar{I}(t)$$

$\bar{I}^{\text{ref}}$ uses public-tape orderFlow; $\bar{I}^{\text{sim}}$ uses the engine's input trade path.

Modelling choice: why AFS sqrt? A constant-rate parent of size Q over T drives the OU recursion to equilibrium $\bar{I}^* = f(Q/T)/\beta$, so

$$\frac{\Delta P}{P_0} \propto f(Q/T).$$

The kernel's shape *propagates directly* to the parent (decay is linear – $1/\beta$ rescale, not reshape):

- **OW** ($f = q$): $\Delta P \propto Q/V$ – linear in size
- **AFS** ($f = \sqrt{q}$): $\Delta P \propto \sqrt{Q/V}$ – the empirical **Almgren / Bouchaud law**

Picking AFS at the engine level wires the empirical regularity in by construction.

Generic trades. Engine accepts any signed schedule via `trade_provider`:

`make_optimal_provider` (OW/AFS) and
`make_fixed_provider`
 (TWAP/VWAP/manual).

Carry modes.

- **daily** – reset $\bar{I}$ + inventory each session
- **multi** – persist with overnight decay $(1 - \beta)^{16^{\text{h}} \cdot 6}$ on $\bar{I}$; **inventory carries regardless**

Stock splits. Heuristic detector flags $|\Delta p|$, $|\Delta v|$ opposite-signed + stable notional. 2019 top-20: **none detected**.

Synthetic alpha (§2.4)

$$\alpha_t = r_t^h + y \frac{dW_t}{P_t}, \quad y = \sqrt{(\rho^{-2} - 1) \frac{\text{Var}(r^h)}{E[P^{-2}]h}}$$

Choosing $x=1$ in $\alpha = x r^h + y dW/P$ gives:

- **Unbiased:** $E[r^h \mid \alpha_t] = \alpha_t$
- **Target correlation:** $\text{Corr}(\alpha, r^h) = \rho$

We use $\rho = 0.05$, $h = 1$ bin (10 s); empirical ρ within $\pm 1\%$ of target.

OW closed form (§2.5)

$$X_t^* = \frac{\alpha_t \text{ADV}}{2\lambda\sigma}, \quad \kappa = \beta = \frac{\ln 2}{H_{\text{bins}}}$$

Position adjusts to X^* at speed κ ; ramped to zero over the final 30 min.

AFS variant. Same target rule per `project.ipynb`; sqrt nonlinearity enters via the simulator's $\tilde{q}$.

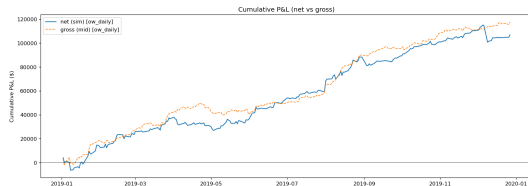
Ext-OW with $\lambda(t)$. **Placeholder.** Simulator already accepts a per-bin λ_t array; HJB-optimal trade rule is not κ -constant when λ varies – closed form pending.

Position trajectory sketch: X_t rises toward X_t^ at exponential speed κ , clips at $\pm 0.5\%$ ADV, ramps linearly to zero in the last 30 min. OW and AFS positions coincide; the cost accounting differs.*

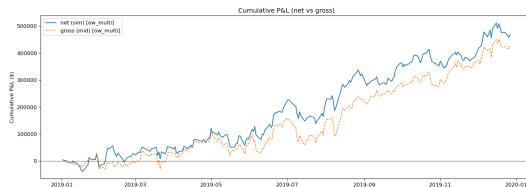
Performance – cumulative P&L

Daily-reset vs multi-day, OW model

§2.6



ow_daily: **Sharpe +3.43**, \$107k net, \$11k impact



ow_multi: Sharpe +2.05, **\$467k net**, impact
–\$39k (carry credit)

Net P&L (blue solid) marks to the Waelbroeck-distorted P_{sim} ; gross P&L (orange dashed) marks to the undistorted mid.

- **Daily-reset wins on Sharpe** – consistent recycling, tight drawdowns (–\$14k vs –\$88k)
- **Multi-day wins on dollars** – $\sim 4\times$ net P&L from accumulated inventory
- Multi-day's net *exceeds* its gross: a long position carried across sessions benefits from its own simulated impact contribution under the Waelbroeck mark-to-market – **the sign convention of "impact cost" needs care** when inventory persists
- Turnover essentially identical between carry modes (\$84M)

TCA decomposition (totals)

§2.6

Net = Gross – Impact, with α rewards separated

Config	Gross P&L	Impact	Net P&L	Pred. α	Real. α	Turnover
ow_daily	116,311	11,787	104,524	44.6 M	116,311	84.0 M
ow_multi	435,399	97,813	337,586	44.6 M	435,399	84.0 M
afs_daily	119,205	56,417	62,787	44.6 M	119,205	86.6 M
afs_multi	482,557	129,999	352,558	44.6 M	482,557	86.6 M

- **Impact cost = Gross – Net** – well-defined under Waelbroeck construction
- **Realised α reward** $\sum_t \text{fwd_ret}_t \cdot X_t \cdot P_t \equiv$ Gross P&L by construction
- **Predicted α reward** $\sum_t \alpha_t \cdot q_t \cdot P_t$ – dominated by α 's noise component at $\rho = 0.05$ (only $\rho^2 = 0.25\%$ of $\text{Var}(\alpha)$ is signal); informative *across* configs, not for level
- Slippage components needing fill-price detail (arrival vs VWAP) intentionally omitted

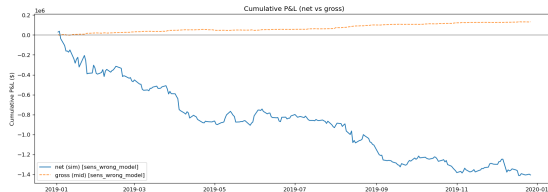
Live sensitivity sweeps (OW, daily-reset)

Scenario	Sharpe	Net P&L
Baseline	+3.43	\$107k
Wrong model (OW $\rightarrow$ AFS)	-2.92	-\$1.41M
Signal delayed 1 min	-0.02	-\$1k
$\rho = 0.01$	+0.62	\$20k
$\rho = 0.10$	+7.37	\$246k
$H = 30$ min	+5.22	\$206k
$H = 120$ min	+2.51	\$58k

Takeaways

- **Wrong model catastrophic** – \$1.41M loss, $\sim$ \$1.54M impact cost
- **1-min delay erases the edge** – Sharpe collapses to ≈ 0 , α is short-horizon
- ρ scales Sharpe linearly:
 $\{0.01, 0.05, 0.10\} \rightarrow \{0.62, 3.43, 7.37\}$
- H is forgiving: $\pm 2\times$ stays in $[+2.5, +5.2]$

Anatomy of the worst case. Fit OW $\hat{\lambda}$ on the panel, then trade as if impact were AFS (sqrt). Year-long net P&L:



Gross alpha-capture (orange dashed) still builds steadily – the signal is real. But OW-sized positions under AFS impact pay $\sim$ \$1.54M in impact cost, dwarfing gross. Net (blue) collapses early and never recovers.

$\Rightarrow$ **Model risk dominates signal risk.**

What we learned

Re-stating the two enhancements with one bullet each

Enhancement 1 – Non-parametric impact with γ regularisation (§2.2)

Apples-to-apples pooled OOS R^2 : **OW** 0.161 $\rightarrow$ 0.221 (+5.9 pp, linearity binding); **AFS unchanged at 0.263** (square-root on the efficient frontier; NP buys nothing). γ shrinkage immaterial in the data-rich headline regime; γ^*/n rises 6 $\times$ and **OOS gain** 14 $\times$ as the training window shrinks 10 $\times$ – the estimator behaves correctly under stress.

Enhancement 2 – Multi-day carry backtest (§2.3)

Carrying $\bar{I}$ and inventory across sessions **quadruples gross P&L** (\$118k $\rightarrow$ \$428k) – accumulated inventory rides the trend. Daily-reset wins on Sharpe (+3.43 vs +2.05) with tight drawdowns; multi-day wins on absolute dollars but introduces a subtle artefact: under Waelbroeck mark-to-market the carried long position benefits from its own simulated price displacement, producing a **negative reported impact cost** (−\$39k). This is an accounting effect, not free money – a clean impact-cost decomposition for multi-day inventory needs a different framing (e.g. implementation-shortfall against an arrival-price benchmark). The engine handles both modes via a single flag.

Honest limitations: overnight α (§2.4 to-expand) and extended-OW $\lambda(t)$ closed form (§2.5 to-expand) are flagged but not implemented; the simulator side of $\lambda(t)$ is already plumbed via `ImpactModel.lam_t_lookup`.

Thank you

Questions?

Code: `src/price_impact/` | Notebook: `final_pipeline.ipynb` | Runs: `saved/<name>/`